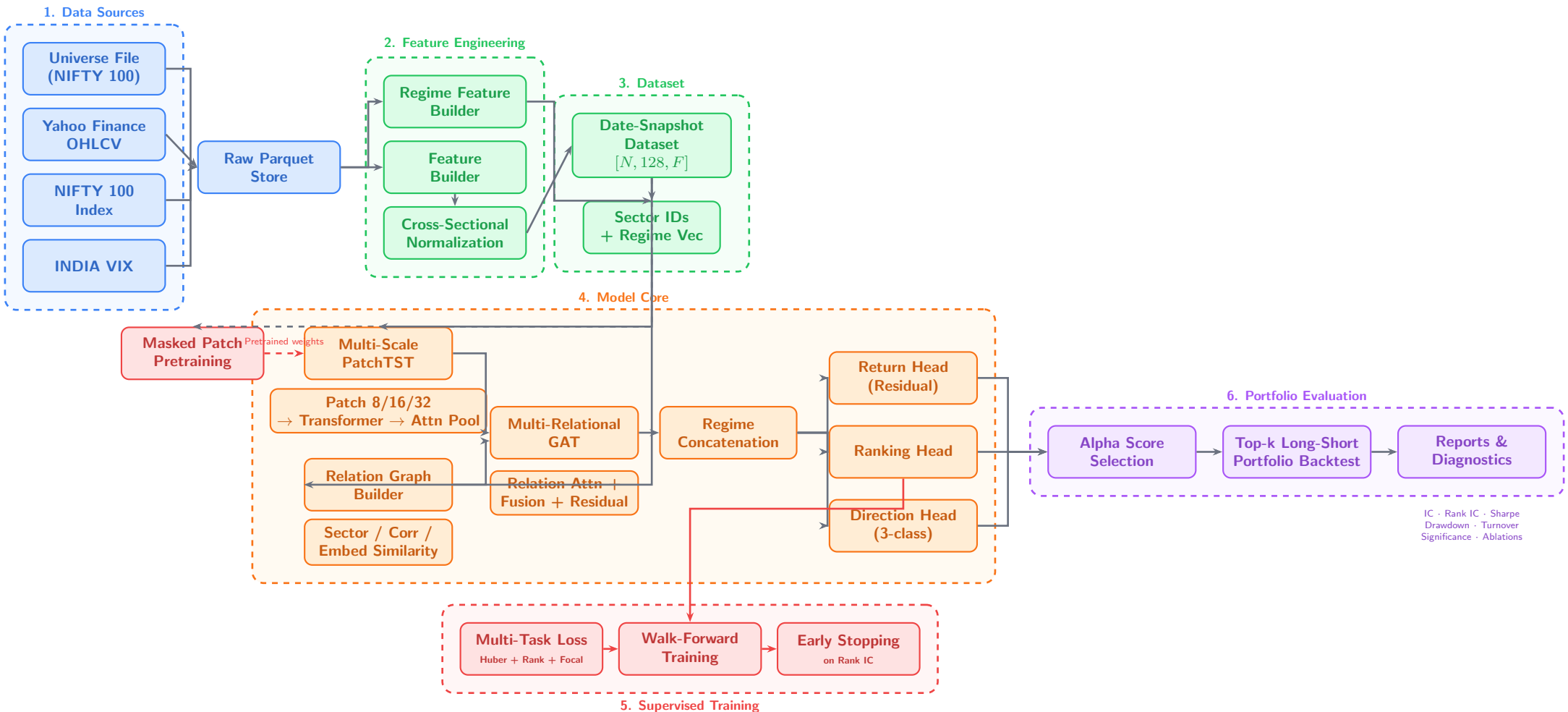




End-to-end architecture of the NIFTY 100 cross-sectional forecasting system. Daily stock, index, and VIX data are converted into cross-sectionally normalized feature tensors. A multi-scale PatchTST encoder generates stock embeddings refined via a multi-relational GAT. Regime-aware multi-task heads predict residual return, ranking score, and direction, evaluated in a walk-forward top-k long-short portfolio framework.